

De Hexagon: A High-Performance Dietary System for Efficient Living

Submitted by

Name: Jeff Sia Yu Heng

Program: Bachelor of Software Engineering (Hons)

"Discipline is the bridge between goals and accomplishment." – Jim Rohn

Table of Content

Cover page	1
Table of Content	2
List of Figure	3
List of Table	4
1.0 Introduction	5
2.0 System Overview	6-10
3.0 Workflow & Operational Details.	11-17
4.0 Cost Control Strategy	18-19
5.0 Long-Term Impact of the Dietary System	20-21
6.0 Conclusion	22
7.0 Reference.	23

List of Figure

Figure 1.0 Organization Chart.....	12
------------------------------------	----

List of Table

Table 1.0 Macronutrient Breakdown Table.	9
Table 2.0 Vitamins & Minerals Table.	10
Table 3.0 Caloric Breakdown Table.	10
Table 4.0 Core Ingredients Table.	13
Table 5.0 Cooking Techniques Table	15
Table 6.0 Multithreading Process Table	16
Table 7.0 Time Allocation Overview Table	17
Table 8.0 Cost Comparison Table	18
Table 9.0 Long-Term Savings Projection Table	19

1.0 Introduction

The starting point of the “De Hexagon” dietary system was not about pursuing a theoretically perfect model. Instead, it was about confronting the inevitable constraints of real life—limited time, tight budgets, fluctuating energy levels, and scarce attention. What I aimed to build was a system that could operate continuously, optimize dynamically, and adapt to ever-changing needs. This system doesn’t merely answer the question “what to eat,” but also addresses *how to eat more systematically, efficiently, and with less burden*.

The core philosophy of the system is inspired by Immanuel Kant’s *Critique of Practical Reason*—where reason is not passive contemplation but an active principle that directs the will toward autonomy and moral self-perfection (*Critique of Practical Reason*, Kant, 1788/1996).. Guided by this principle, I materialized “practical reason” into a structured daily eating system: it emphasizes smooth execution, minimized behavioural resistance, and systematized planning to alleviate decision fatigue. In short, this is an attempt to turn rationality from an abstract ideal into a tangible lifestyle system.

“De Hexagon” is not a collection of recipes, but a life-strategy model encompassing time efficiency, nutritional density, taste adaptability, workflow automation, and cost control. Every decision—from ingredient purchasing, pre-processing, cooking methods to portioning logic—has been optimized and tested to ensure the system is both efficient and sustainable.

This report comprehensively outlines the background, operation flow, nutritional structure, optimization outcomes, and theoretical support of the system—showcasing a model that sits between personal experimentation and a lifestyle philosophy. My goal is simple: to embed rationality not only in thought but in every action and choice of daily life.

2.0 System Overview

2.1 System Objectives

- **Timesaving:** Reduce daily meal prep time
- **Cost control:** Keep weekly food budget under RM 40
- **Nutritional optimization:** High protein, sufficient vegetables, moderate carbs
- **Process automation:** Simplify decisions and operational steps
- **Taste management:** Avoid flavour fatigue and increase meal enjoyment
- **Cognitive load reduction:** Standardized execution, minimize distractions

2.2 Core Components

The **Hexagonal Combat Meal** system is built upon four interlocking components that ensure efficiency, structure, and sustainability in daily dietary management:

- **Ingredient Management System**

This component ensures that all ingredients are categorized, tracked, and replenished in a systematic manner. By using an inventory-based approach, it reduces food waste, maintains consistency in meal quality, and supports meal planning aligned with nutritional goals.

- **Cooking and Storage Workflow Optimization**

Meals are prepared in batches using a streamlined cooking process that minimizes redundant steps and maximizes efficiency. Proper portioning, labeling, and containerization ensure that meals are stored safely and remain fresh, facilitating quick access and consumption throughout the week.

- **Budget and Cost Control Mechanism**

By standardizing meals and sourcing ingredients in bulk, this mechanism allows for significant cost savings while maintaining nutritional quality. Every purchase is evaluated for its cost-to-value ratio, promoting a lifestyle that is both financially sustainable and health-conscious.

2.3 System Evolution:

From ABC Soup to the Hexagonal System The prototype of “De Hexagon” emerged during my university years as a self-experiment to solve real issues: expensive meals, poor nutrition, and the hassle of cooking. The system wasn’t born out of idealism—it started with a bowl of soup, refined step by step.

2.3.1 Starting Point: ABC Soup Experiment

The earliest signature dish was the frequently made “ABC Soup” in our dorm.

“ABC Soup is a traditional Malaysian home-style soup made from potatoes, carrots, corn, onions, and chicken. It’s balanced and mild in flavour.”

However, we encountered two main issues during execution:

1. **Chicken preparation was too troublesome:** Defrosting, cleaning, chopping—too time-consuming
2. **Lack of storage tools:** Daily ingredient handling was exhausting and easy to give up

So, we conducted an ultra-simple experiment:

For a month, we eat “vegan ABC soup” with **only corn, carrot, potato, and onion**.

Our sole protein source was **two boiled eggs per day** and Carbohydrates with **rice**.

While this method saved time and filled our stomachs temporarily, after a few weeks we noticed changes in our physical and mental states:

- Easy fatigue and decreased concentration
- Psychological boredom due to repetitive meals
- Severe lack of protein and calories

2.3.2 Nutritional Adjustment

To improve meal quality, I decided to **bring chicken back**, but in a more efficient way:

- Bulk-marinated chicken + portioned freezing for storage
 - Purchase large quantities of chicken breast or thigh
 - Clean, trim, and dice, then marinate with soy sauce, pepper, and garlic powder
 - Portion into **250g bags**, freeze, and use as needed

2.3.3 Advanced Phase: Batch Prep and Systematization of Chicken

Despite chicken being essential, prep work remained tedious. So, in a new semester, I developed a more practical strategy:

“Batch marination + portioned freezing + modular ingredient prep”

Here’s how it works:

- Purchase large quantities of chicken
- On prep day, clean, cut, and marinate in bulk
- **Portion 250g each** into plastic bags and freeze
- Prepped vegetables are stored by type in sealed containers in the fridge

This turned daily cooking from “decision-making + raw ingredient prep” into **a streamlined, stable, and sustainable workflow.**

2.4 Nutritional Value Analysis

The core strength of the “Hexagonal Battle Meal” lies in its **well-structured, balanced, and clearly organized meal model**, ideal for students seeking health and efficiency within tight time and budget constraints.

2.4.1 Macronutrient Breakdown

Module	Main Ingredients	Nutrient Contribution
Chicken (250g)	Chicken breast or thigh	High protein, low fat ($\approx$ 55g protein / 300 kcal)
Rice (1 bowl)	Cooked rice $\approx$ 150g	Main carb source ($\approx$ 45g carbs / 200 kcal)
ABC Soup Veg	Potato, carrot, corn, onion	Carbs, fiber, vitamins ($\approx$ 20g carbs, 5g fiber)

Table 1.0 Macronutrient Breakdown Table

Protein:

- $\approx$ 55g protein per meal — supports muscle and immunity
- Chicken is a high-quality animal protein, well-absorbed, moderate fat

Carbs:

- Rice + potato + corn = slow-releasing carb structure
- Avoids blood sugar spikes, increases satiety and energy duration

Fat:

- Chicken breast = low fat, helps with body fat control
- Optional healthy oils (olive, sesame) enhance flavour

2.4.2 Vitamins & Minerals

Nutrient	Source Vegetables	Function
Vitamin A	Carrot	Eye health, immune support
Vitamin C	Onion, corn	Antioxidant, aids iron absorption
Vitamin B	Corn, potato	Energy metabolism, nerve system protection
Potassium	Potato	Electrolyte balance, cardiovascular support
Iron	Chicken + vegetables	Haemoglobin synthesis, prevents anaemia
Dietary Fiber	All vegetables	Gut health, improves digestion

Table 2.0 Vitamins & Minerals Table

2.4.3 Caloric Breakdown (Estimates)

Module	Estimated Calories (kcal)	Notes
Chicken	300	250g chicken breast/thigh
Rice	200	1 bowl cooked rice ($\approx$ 150g)
Veg Soup	120	Potato, carrot, corn mix
Total	620 kcal	Ideal for a balanced meal

Table 3.0 Caloric Breakdown Table

Two of these meals per day + one light lunch or snack = $\sim$ 1600–1800 kcal/day, suitable for college students with moderate physical and mental activity.

3.0 Workflow & Operational Details

3.1 Core Execution Trio Introduction

Behind the high-intensity, high-efficiency kitchen system stands a trio of key players, each with their own role but working in close coordination. They not only uphold daily operations but also transform each meal into a fusion of efficiency and artistry.

3.1.1 Team Core Member

Jeff Sia Yu Heng

Chief Operations Originator (COO²) & Senior Ingredient Transformation Engineer

- Architect of the system; leads planning, optimization, and execution
- Specializes in ingredient procurement, cost control, and workflow design
- Oversees team coordination and ensures smooth, efficient operations
- Manages food prep and drives team morale

Lim Chun Xin

Culinary Strategy Executive Officer (CSEO)

- Leads cooking operations with precision and consistency
- Handles ingredient transport and prepares all key dishes (rice, vegetables, protein)
- Ensures high-quality output as the sole hot kitchen operator

Vernon Moo Jing Rong

Director of Dish Integrity

- Maintains hygiene, safety, and post-meal cleanup
- Responsible for dishwashing, sanitation, and tool storage
- Ensures a clean and organized kitchen environment behind the scenes

3.1.2 Organization Chart

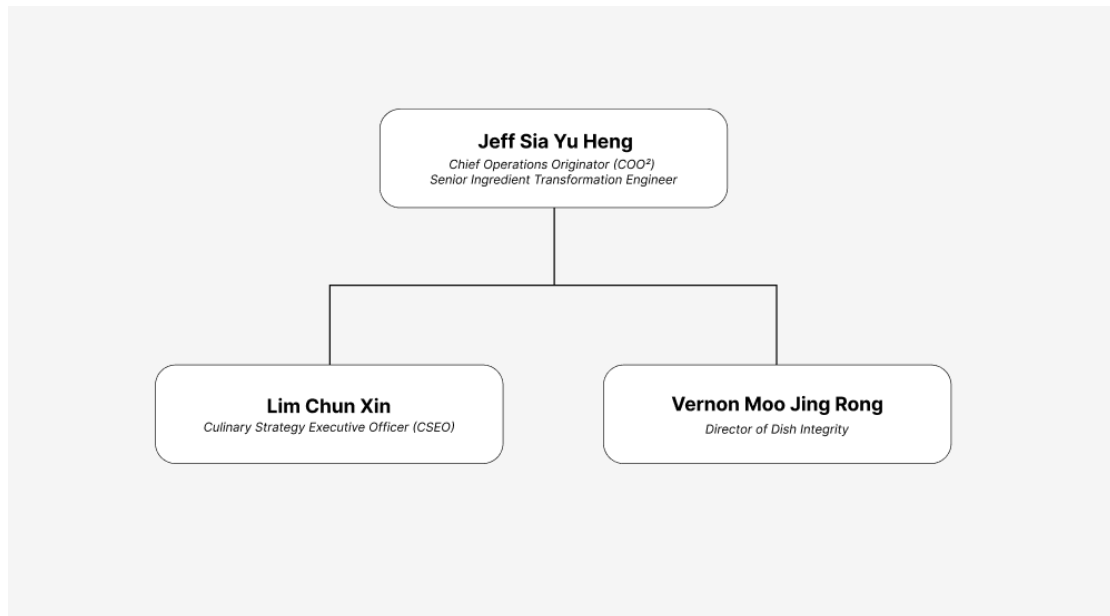


Figure 1.0 Organization Chart

3.2 Ingredient Types and Management

3.2.1 Core Ingredients

Category	Ingredients	Characteristics
Protein	Chicken breast, eggs	High-protein, low-fat, easy to store
Carbs	Potatoes, carrots, corn	High-fiber, stable blood sugar, slow energy release
Vegetables	Cabbage, onions, broccoli, etc.	Rich in vitamins, antioxidants

Table 4.0 Core Ingredients Table

3.2.2 Ingredient Processing Workflow

- **Procurement frequency:** Twice weekly, bulk purchase
- **Pre-processing steps:**
 - Chicken is marinated by flavour (e.g., original, spicy), vacuum-sealed, and frozen.
 - Vegetables (potatoes, carrots, corn, cabbage) are washed, chopped, and stored refrigerated or frozen.
 - Eggs are kept fresh and used within a short period as needed.

3.2.3 Storage Management

Inventory rotation with First-In-First-Out (FIFO) principle to minimize waste:

- Earlier-purchased items are used first to ensure freshness.
- Frozen items (like chicken) are dated and thawed in order.
- Fresh vegetables are prioritized based on how long they've been stored to avoid spoilage or waste.

Why is this important?

FIFO not only reduces food waste and controls budget but also ensures nutritional freshness, further improving the quality and efficiency of the dietary system.

How to integrate into the system?

After each grocery trip:

- Sort and organize ingredients by date
- Store dry goods like potatoes and onions by batch or label
- During meal prep, always pick the oldest stock first—automation by habit

3.3 Cooking Methods and Optimization

3.3.1 Cooking Techniques

Ingredient	Method	Purpose
Chicken	Air frying	Low-fat, high-protein, hands-free cooking
Vegetables	Bulk boiling	Softens fibre, retains nutrients, aids digestion
Mixed meals	Rice bowl mix	Balanced texture, layered flavour, full nutrition

Table 5.0 Cooking Techniques Table

3.3.2 Taste and Texture Management

- Rotate marinades (e.g., black pepper, garlic, spicy) to avoid taste fatigue.
- Add appropriate salt and MSG during boiling to enhance natural flavour
- Maintain some crunch in vegetables—avoid overcooking for better texture.

3.4 Time Management Strategy

One of the core features of the “Hexagonal Combat Meal” system is **efficient meal prep**. Through **batch preparation + multithreaded thinking**, it minimizes daily time investment and allows a busy student lifestyle to coexist with healthy eating.

3.4.1 Daily Meal Prep Process (Multithreading Model)

This model incorporates the concept of parallel tasking—such as preparing rice, steaming vegetables, and marinating proteins simultaneously—optimizing kitchen time and reducing decision fatigue. Inspired by principles of time blocking and multitasking efficiency, it transforms meal prep into a streamlined and scalable daily routine (Prajapati et al., 2025).

1. Thaw chicken in advance

- Chicken must be transferred from freezer to fridge **8 hours in advance**.
- Recommended: Place it in the fridge the night before for next-day use.

2. Initiate cooking threads (parallel processing)

- Execute these three tasks concurrently:

Thread	Task	Duration
Thread A	Rice cooking (automatic rice cooker)	~25 minutes
Thread B	Vegetables + chicken (air fry/boil)	~15–30 minutes
Thread C	Eggs (boil in room temp water)	7–8 minutes

Table 6.0 Multithreading Process Table

Eggs should be boiled starting in room temperature water to ensure even cooking and a soft, runny yolk.

3.4.2 Time Allocation Overview

Phase	Manual Time	Total Duration (incl. waiting)
Thawing	None	8 hours
Actual Prep	< 35 mins	~35 minutes
Workload	Very low (mostly automated)	

Table 7.0 Time Allocation Overview Table

3.4.3 Multithreading Theory Support

Applying the concept of multithreading from computer science to real life:

In computing, multithreading allows several tasks to execute “simultaneously” to optimize resource use and reduce idle time. In the “Hexagonal Combat Meal” system, this concept is adapted to daily life:

- **Cookware = Core resources (CPU)**
- **Rice, vegetables, eggs = Threads**
- **Human action = Main thread scheduling**

With parallel execution, we:

- **Minimize meal prep time**
- **Maximize productivity per time unit**
- **Improve the system's "throughput" and "real-time responsiveness"**

4.0 Cost Control Strategy

4.1 Weekly Cost Overview (Based on 10 Meals per Week)

Through a systemized approach h to ingredient sourcing, meal preparation, and food storage, this dietary management system achieves highly optimized cost efficiency while maintaining nutritional integrity. By applying principles from linear programming and strategic meal planning, the system balances affordability and nutrition — a result supported by studies showing that optimized diets can be both healthier and cheaper when structured deliberately (arXiv, 2024).

Additionally, the adoption of simplified meal planning models — such as the "6-to-1 Grocery Method," which limits unnecessary purchases — and meal prepping routines reduces food waste and impulse spending (Business Insider, Wikipedia: Meal Preparation).

(1) Systemized Meal Budget (Per Week)

- Weekly cost for 10 meals:
 $10 \times \text{RM}4 = \text{RM}40/\text{week}$
- Each meal meets basic macronutrient goals: high protein, sufficient vegetables, and controlled carbohydrates.

(2) Typical Outside Food Budget

- Average cost per meal when eating out: **RM12.50**
- Weekly cost for 10 meals:
 $10 \times \text{RM}12.50 = \text{RM}125/\text{week}$

(3) Cost Comparison Table

Category	Weekly Budget (RM)	Cost Difference (RM)	Cost Reduction (%)
Systemized Meal Plan	40	-	-
Eating Out	125	+85	-

Table 8.0 Cost Comparison Table

Weekly **Savings**: RM85

Cost **Reduction Percentage**:

$$(85 \div 125) \times 100 = 68\%,$$

4.2 Long-Term Savings Projection (Based on 10 Meals/Week)

The following projections illustrate the **cumulative financial benefits** of maintaining a systemized meal plan over time:

- **Monthly Savings:**

$$\text{RM}85 \times 4 = \text{RM}340/\text{month}$$

- **Annual Savings:**

$$\text{RM}340 \times 12 = \text{RM}4,080/\text{year}$$

Time Period	Accumulated Savings (RM)
Monthly	340
Annually	4,080
Five Years	20,400

Table 9.0 Long-Term Savings Projection Table

5.0 Long-Term Impact of the Dietary System

5.1 Physical Optimization

From the perspective of Nutritional Intervention Science, a structured dietary system effectively reduces chronic inflammation markers (e.g., CRP, IL-6), which contributes significantly over time to improved cognition, focus, and reduced anxiety levels (Calder, 2020; Slavin, 2013).

- **High-Protein, Low-Fat, Low-Sodium Diet**

According to nutritional research, protein intake stabilizes blood glucose and supports muscle repair, while a diet low in oil and sodium eases the burden on internal organs, enhancing long-term stamina and sleep quality (Micha et al., 2017).

- **High Fiber Intake (e.g., corn, carrots, potatoes, onions)**

Fiber stimulates gut motility and supports healthy microbiota balance, improving digestion and nutrient absorption. This counters common "sub-health" issues among students, such as constipation and indigestion.

- **Higher Nutrient Density Compared to Fast Food**

The "Hexagonal Power Meal" is built on whole, minimally processed foods, avoiding trans fats, empty calories, and preservatives typical of fast food. This supports long-term mental clarity and emotional stability.

5.2 Enhanced Thinking and Decision-Making Efficiency

Referencing Daniel Kahneman's Dual-System Theory in behavioural economics, standardizing meal routines shifts dietary decisions from System 2 (deliberate thinking) to System 1 (automatic responses). This frees cognitive resources for learning, creative tasks, and strategic work (Kahneman, 2011).

- **Elimination of Daily “What to Eat” Decisions**

According to the **Decision Fatigue Theory**, even minor daily choices (like selecting meals) deplete executive function, reducing our ability to perform more important tasks effectively.

- **Routine and Standardization of Meal Workflow**

Creating a consistent process (e.g., pre-prep → parallel cooking → fixed pairings) reduces behavioural friction, converting actions into habits—what neuroscience terms “automatic behaviour loops.” This minimizes the drain on self-control and enhances productivity.

6.0 Conclusion

The “De Hexagon” dietary management system exemplifies how high-performance thinking can be embedded into daily routines through structured planning, workflow optimization, and strategic resource coordination. Far beyond a simple meal plan, it represents a deliberate operational framework that enhances physical health, sustains mental clarity, and supports time and cost efficiency in a consistent, scalable manner (Micha et al., 2017; Kahneman, 2011).

By automating decision-making and reducing cognitive load, the system reflects principles from behavioural economics and executive function theory—demonstrating how standardization and habit-formation can preserve mental energy for more cognitively demanding tasks (Baumeister et al., 1998; Kahneman, 2011). Moreover, its cost-effective approach aligns with studies showing that nutritionally optimized diets are not only healthier but can also be more affordable when planned intentionally (Maillot et al., 2010).

Ultimately, this system is not merely about nutrition—it is a philosophy of life design. It fuses self-discipline, rational execution, and team synergy into a practical model that transforms everyday choices into expressions of intentional living. In doing so, it redefines food preparation from a routine chore into a vehicle for personal autonomy, growth, and sustainability.

7.0 Reference

Baumeister, R.F., Bratslavsky, E., Muraven, M. and Tice, D.M., 1998. *Ego depletion: Is the active self a limited resource?* Journal of Personality and Social Psychology, 74(5), pp.1252–1265.

Bennett, P., 2025. *I save money on groceries using '6-to-1' method; makes shopping easy*. [online] Business Insider. Available at: https://www.businessinsider.com/how-save-money-groceries-using-6-to-1-method?utm_source=chatgpt.com [Accessed 29 Apr. 2025].

Kahneman, D., 2011. *Thinking, fast and slow*. London: Penguin Books.

Kant, I., 1996. *Critique of practical reason*. Translated by M.J. Gregor. Cambridge: Cambridge University Press. (Original work published 1788)

Maillot, M., Darmon, N., Darmon, M., Lafay, L. and Drewnowski, A., 2010. *Nutritional quality of foods and food choices in French adults: a study from the SUVIMAX cohort*. American Journal of Clinical Nutrition, 91(5), pp.1220–1226.

Micha, R., Peñalvo, J.L., Cudhea, F., Imamura, F., Rehm, C.D. and Mozaffarian, D., 2017. *Association between dietary factors and mortality from heart disease, stroke, and type 2 diabetes in the United States*. JAMA, 317(9), pp.912–924. Available at: <https://doi.org/10.1001/jama.2017.0947> [Accessed 29 Apr. 2025].

Prajapati, U., Jain, T., Machiraju, A. and Kaushik, D., 2025. *Linear Optimization for the Perfect Meal: A Data-Driven Approach to Optimising the Perfect Meal Using Gurobi*. [online] arXiv.org. Available at: https://arxiv.org/abs/2501.04143?utm_source=chatgpt.com [Accessed 29 Apr. 2025].

Slavin, J.L., 2013. *Dietary fiber and body weight*. Nutrition, 29(3), pp.498–505. Available at: <https://doi.org/10.1016/j.nut.2012.11.007> [Accessed 29 Apr. 2025].

Wikipedia Contributors, 2024. *Meal preparation*. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Meal_preparation [Accessed 29 Apr. 2025].